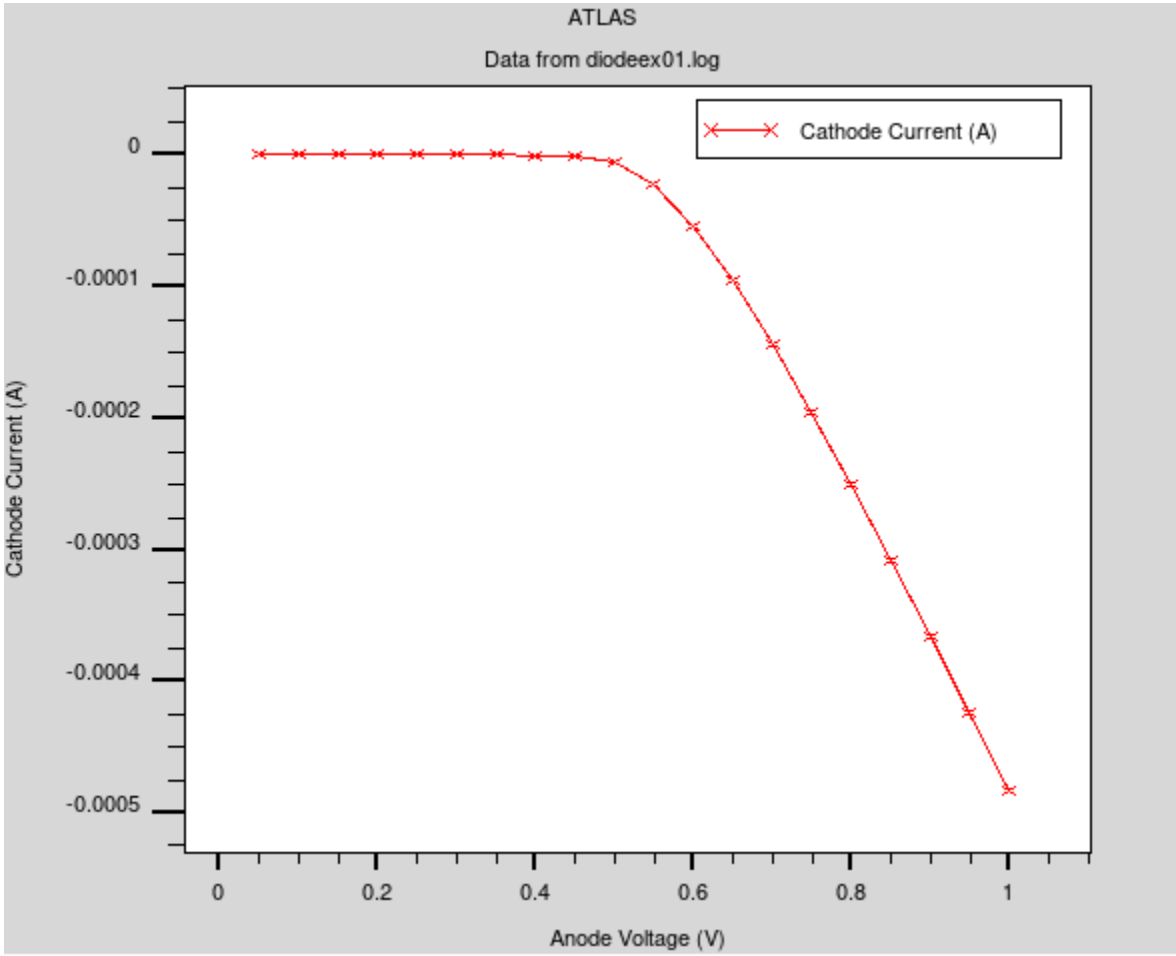
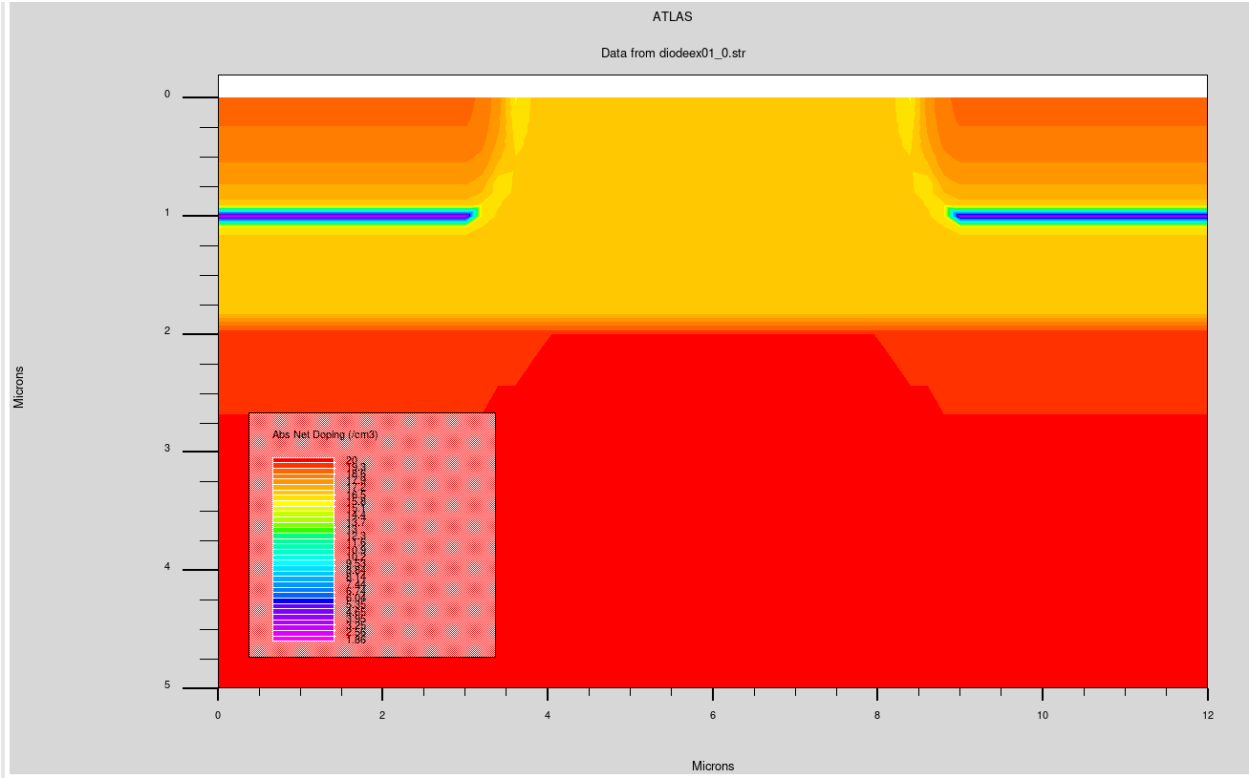
Q3e)

Pro/Con Table

BJT	HBT
- Slower switching time - Slower transit time +Less complicated of a structure -Simpler and perhaps very slightly cheaper to produce -Lower quality factors/efficiency	**+ Importantly has a faster switching time (can switch at higher frequency)** + Higher transit time + Higher efficiency and gain (better quality factors) - More complicated of a structure + More complicated to analyze

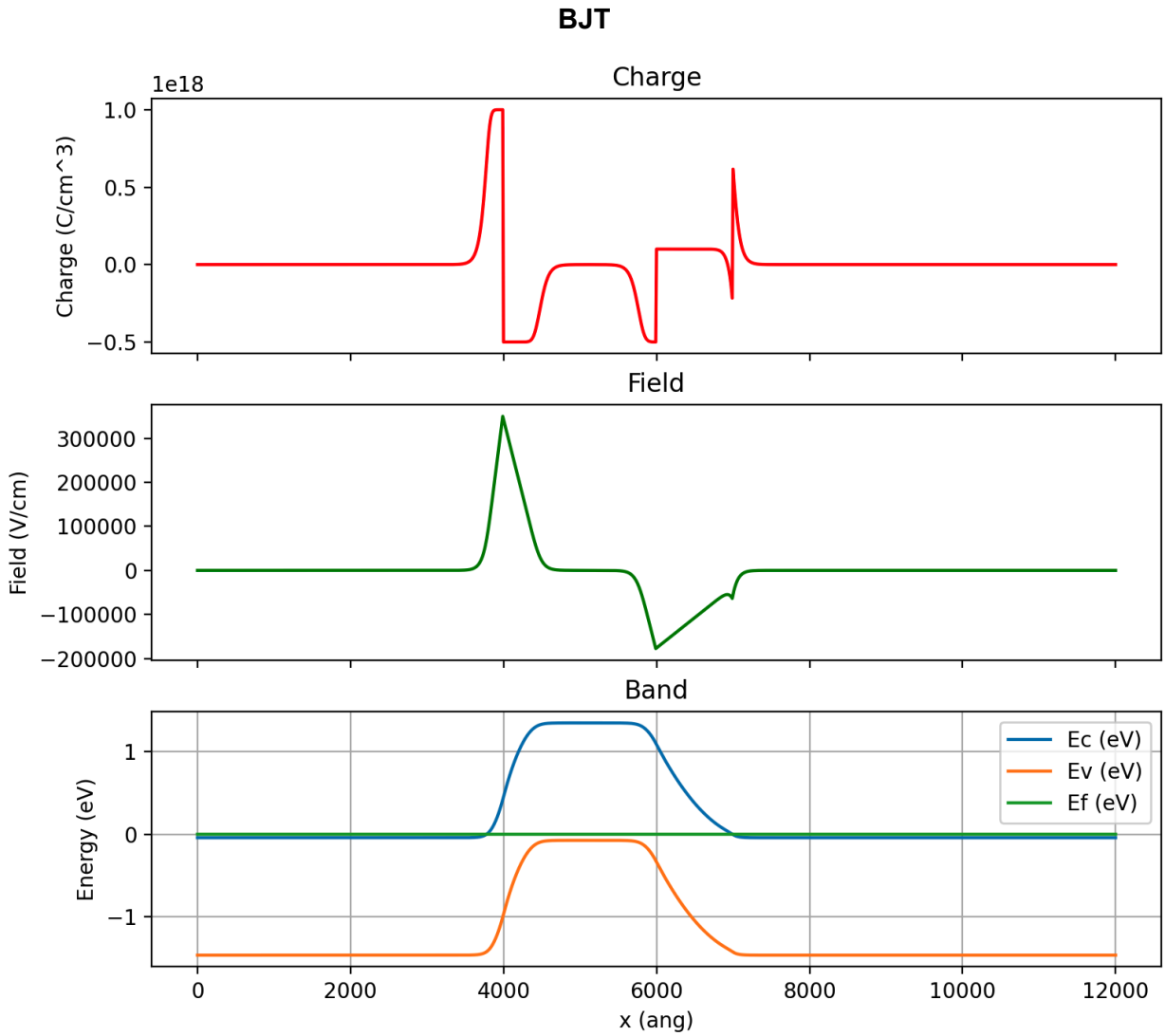
Q3f)
Major players - Intel, IBM, and STMicroelectronics
Applications - automotive radar systems, advanced analog circuits, and satellite communications
Paragraph - After looking online and perusing a few papers and articles, I've noticed that Silicon-germanium heterojunction bipolar transistors are starting to gain in popularity since they offer a lot of useful capabilities. In particular, they are used for devices in and integration with BiCMOS technology. This combination offers significant advantages for high-speed applications, especially in telecommunications, such as 5G and future 6G networks. SiGe HBTs enable efficient performance in power amplifiers and low-noise amplifiers, providing a cost-effective solution without sacrificing performance. The ability of SiGe HBTs to operate across a broad frequency spectrum enhances their versatility, making them essential components in modern electronic systems and paving the way for future innovations in high-frequency technologies. These are all resultant from HBT ideas about mixing existing technologies to create a more robust transistor that can accommodate a wider array of frequencies. For instance, SiGe-based HBTs draw from traditional HBT concepts by incorporating germanium into the silicon structure. This addition enhances carrier mobility, allowing these transistors to operate at higher frequencies. The result is a technology that significantly boosts performance in high-speed applications.

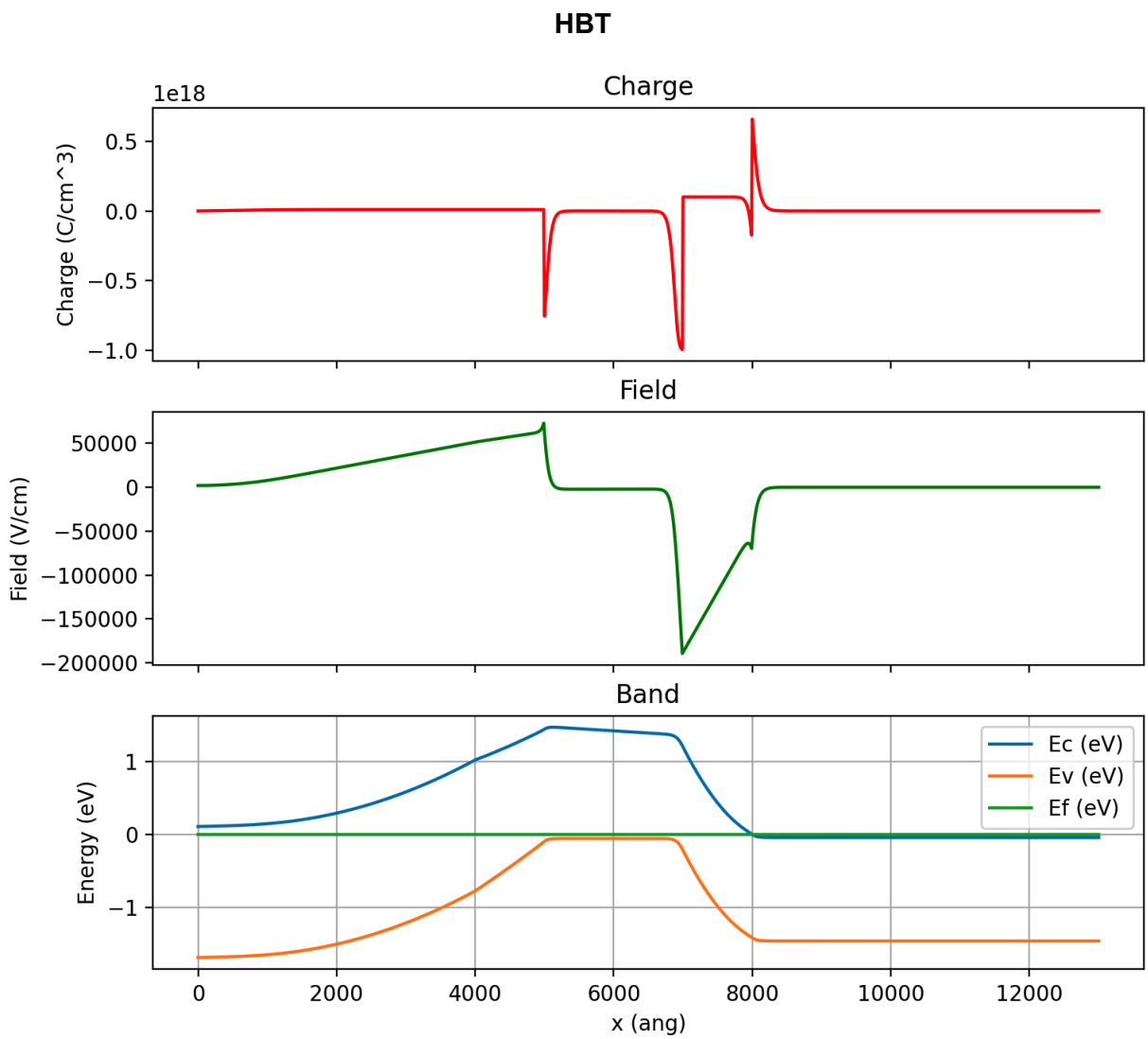

ATLAS
Data from diodeex01.log
Cathode Current (A)
0
-0.0001
-0.0002
-0.0003
-0.0004
-0.0005
0
0.2
0.4
0.6
0.8
1
Anode Voltage (V)
Cathode Current (A)


ATLAS
Data from diodeex01_0.str
Abs Net Doping (/cm3)
20
18.9
17.8
16.7
15.6
14.5
13.4
12.3
11.2
10.1
9.04
8.01
7.41
6.81
6.21
5.61
5.01
4.41
3.81
3.21
2.61
2.01
1.41
0.81
0.21
-0.39
-0.99
-1.59
-2.19
-2.79
-3.39
-3.99
-4.59
-5.19
-5.79
-6.39
-6.99
-7.59
-8.19
-8.79
-9.39
-9.99
Microns
0
1
2
3
4
5
0
2
4
6
8
10
12
Microns

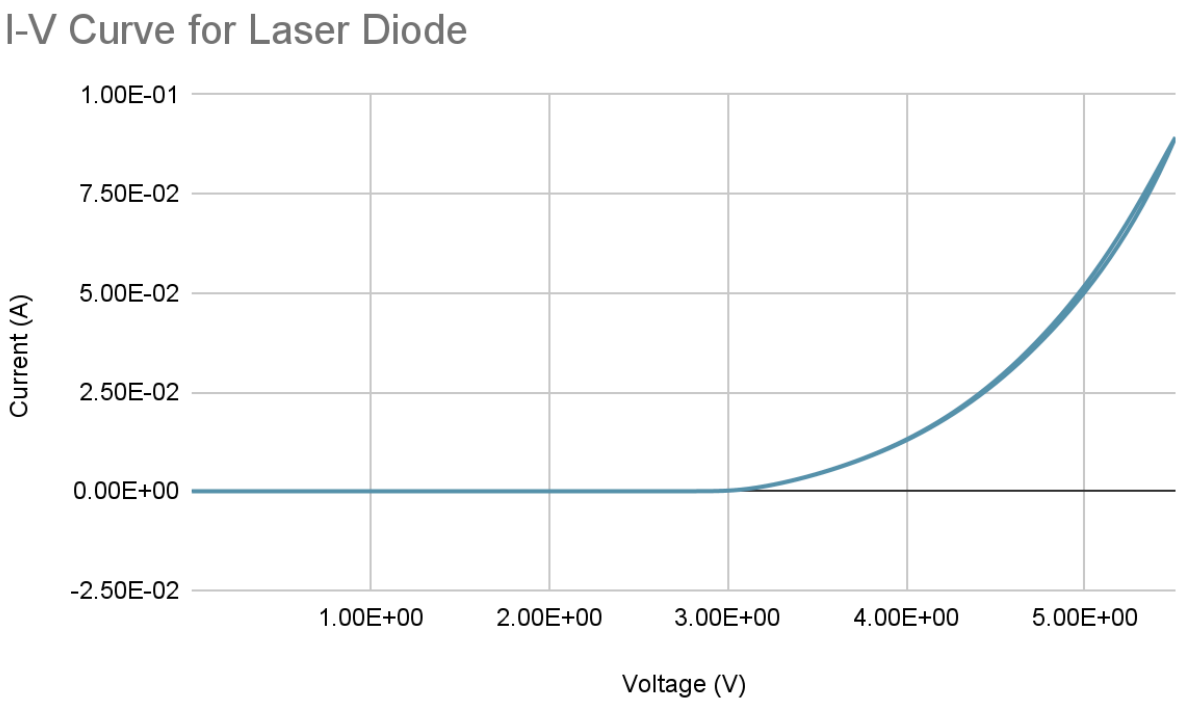
Q3

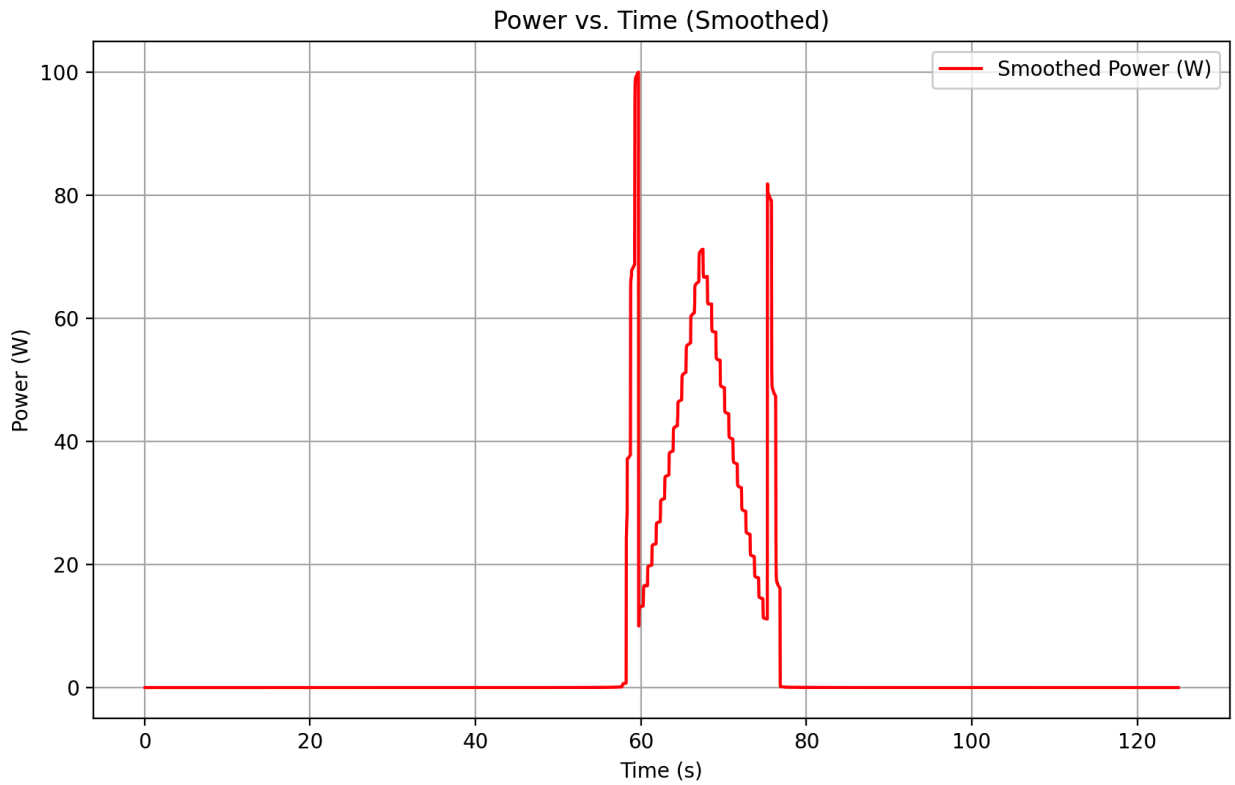
1D Poisson Plots for Q3



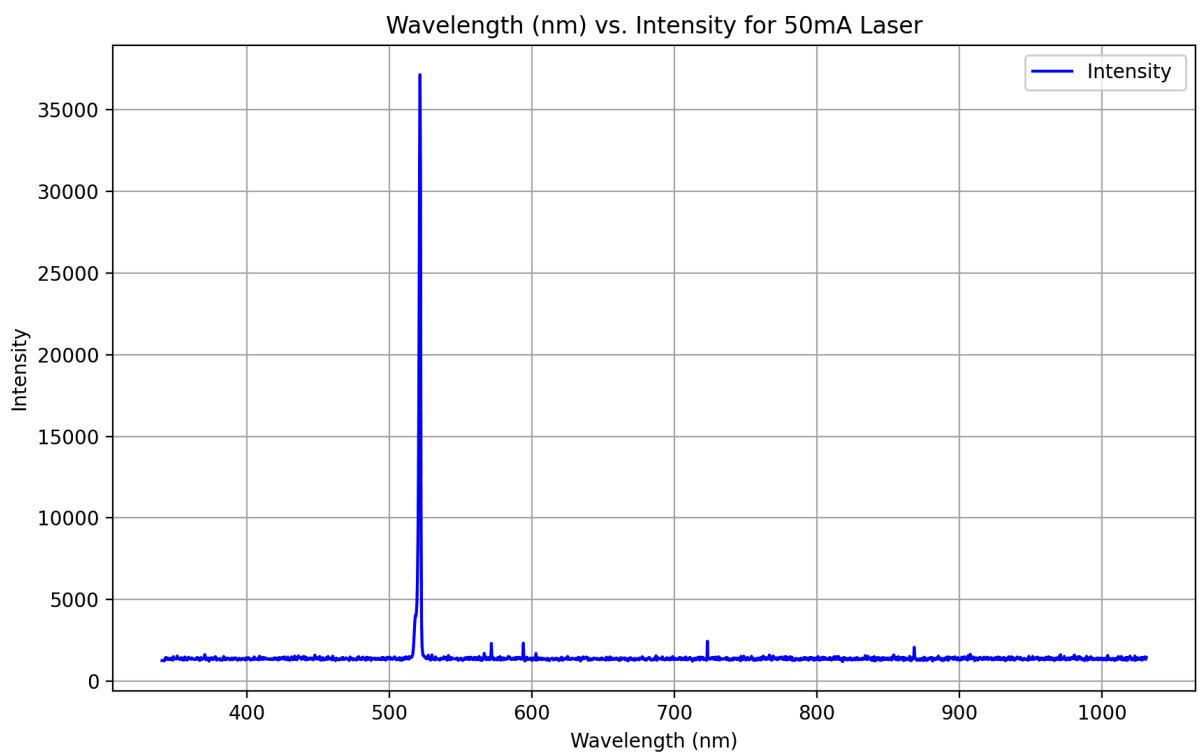

HBT
Charge
1e18
Charge (C/cm^3)
0.5
0.0
−0.5
−1.0
Field
Field (V/cm)
50000
0
−50000
−100000
−150000
−200000
Band
Energy (eV)
1
0
−1
Ec (eV)
Ev (eV)
Ef (eV)
0
2000
4000
6000
8000
10000
12000
x (ang)

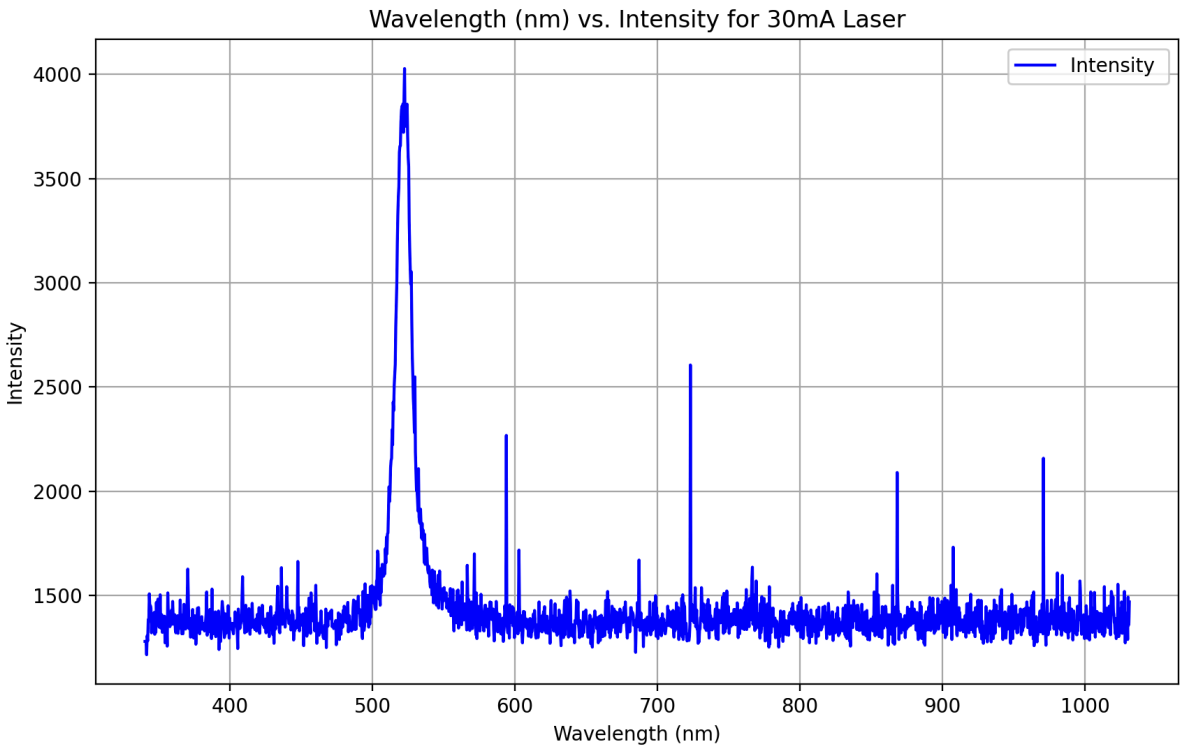
Lab


I-V Curve for Laser Diode
1.00E-01
7.50E-02
5.00E-02
2.50E-02
0.00E+00
-2.50E-02
Current (A)
1.00E+00
2.00E+00
3.00E+00
4.00E+00
5.00E+00
Voltage (V)



You can also notice a slight hysteresis effect from the generated heat when decreasing the voltage after first increasing it to 5.5V.




Wavelength (nm) vs. Intensity for 30mA Laser
Intensity
4000
3500
3000
2500
2000
1500
400
500
600
700
800
900
1000
Wavelength (nm)
Intensity